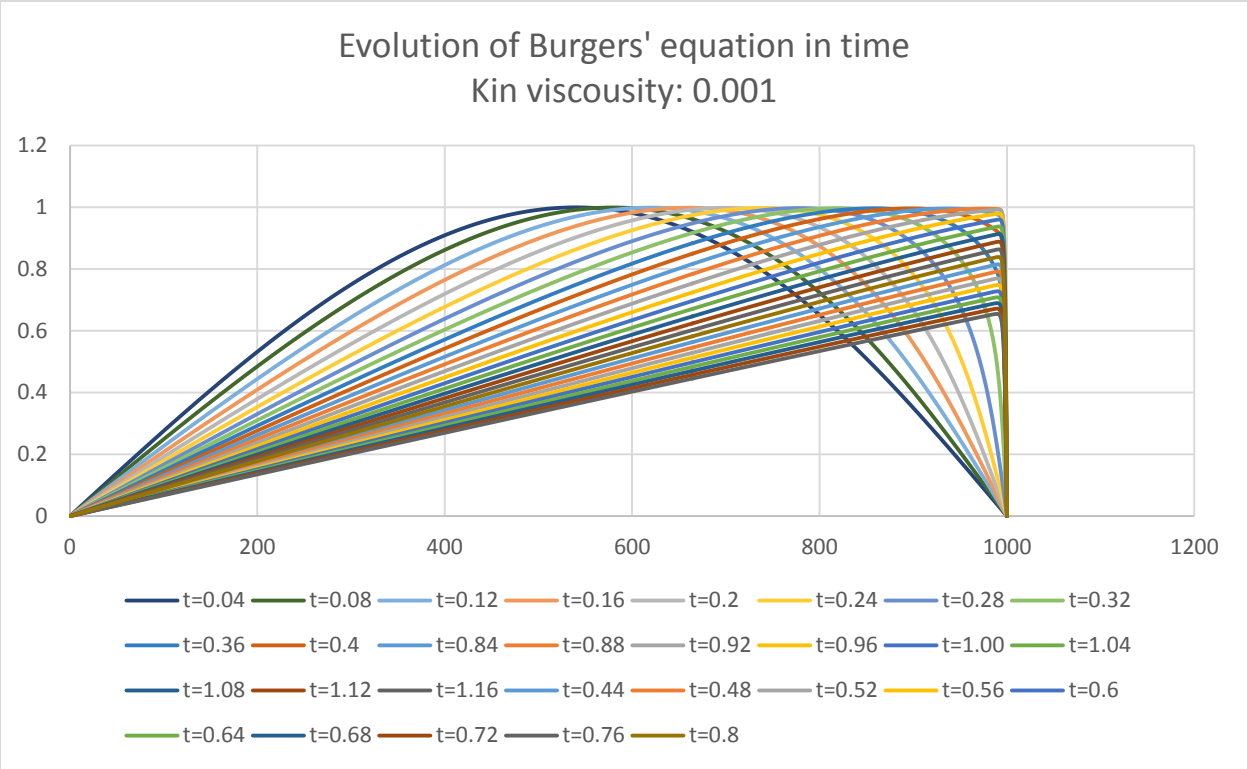
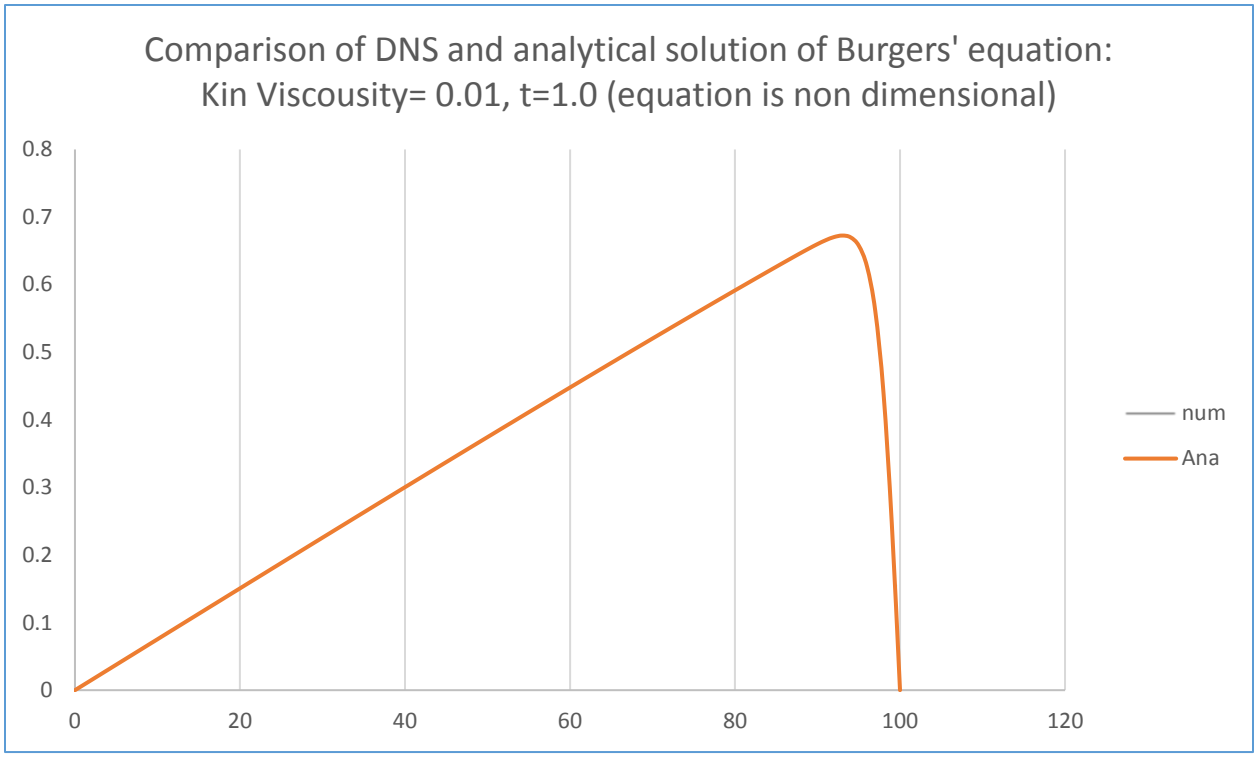
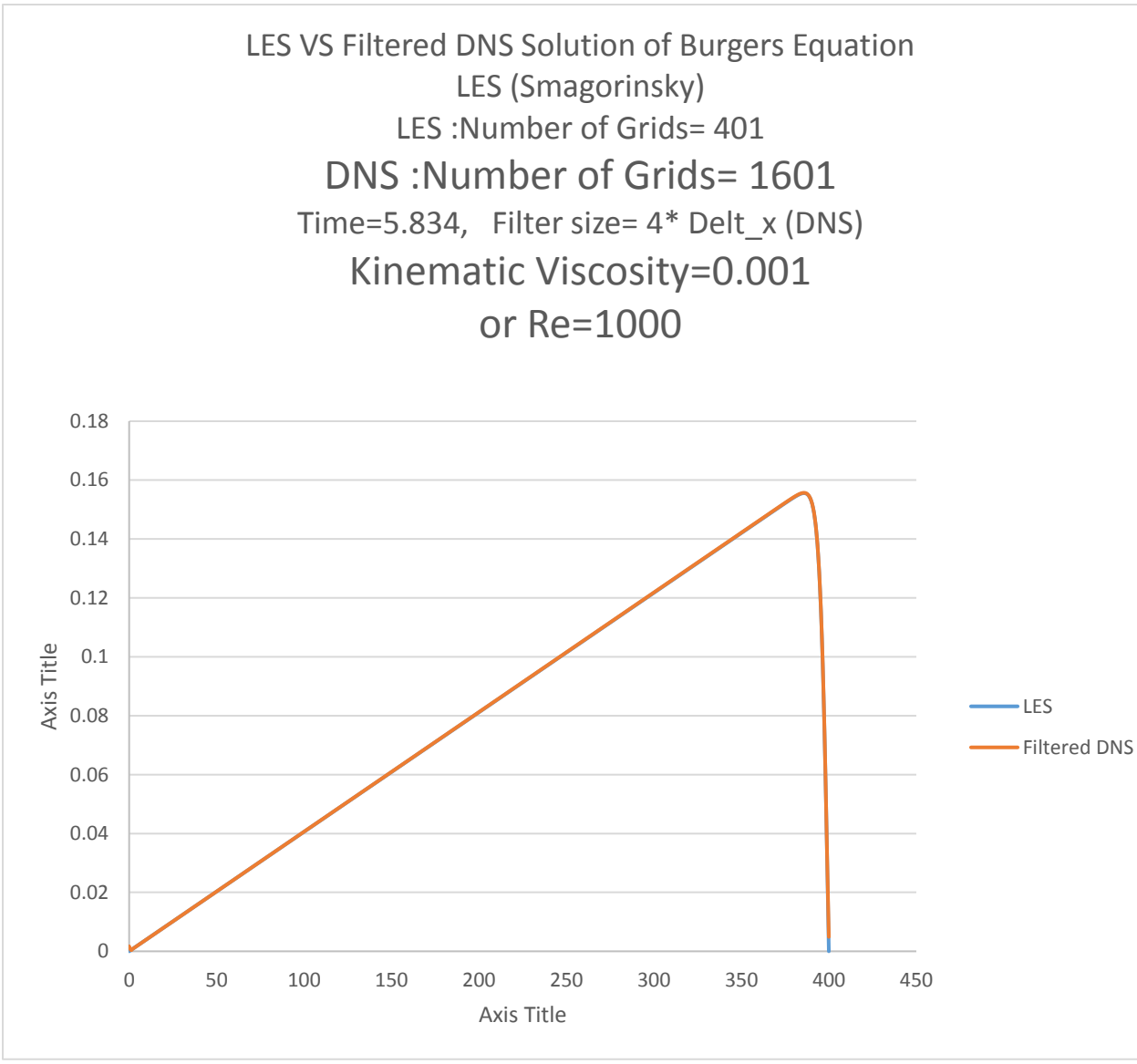
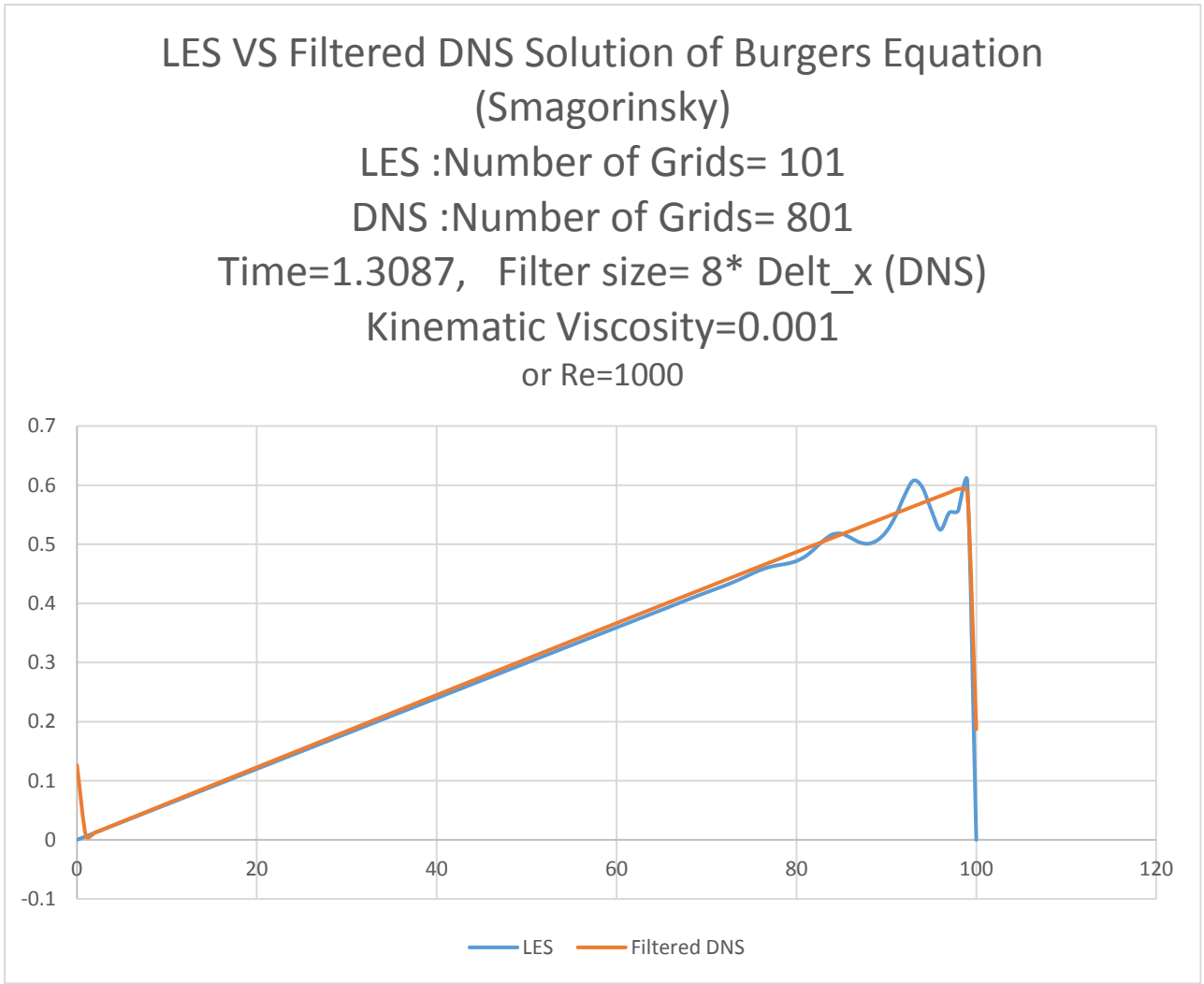

Evolution of Burgers' equation in time
Kin viscousity: 0.001
1.2
1
0.8
0.6
0.4
0.2
0
0
200
400
600
800
1000
1200
t=0.04
t=0.08
t=0.12
t=0.16
t=0.2
t=0.24
t=0.28
t=0.32
t=0.36
t=0.4
t=0.84
t=0.88
t=0.92
t=0.96
t=1.00
t=1.04
t=1.08
t=1.12
t=1.16
t=0.44
t=0.48
t=0.52
t=0.56
t=0.6
t=0.64
t=0.68
t=0.72
t=0.76
t=0.8


Comparison of DNS and analytical solution of Burgers' equation:
Kin Viscousity= 0.01, t=1.0 (equation is non dimensional)
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
20
40
60
80
100
120
num
Ana


LES VS Filtered DNS Solution of Burgers Equation
LES (Smagorinsky)
LES :Number of Grids= 401
DNS :Number of Grids= 1601
Time=5.834, Filter size= 4* Delt_x (DNS)
Kinematic Viscosity=0.001
or Re=1000
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
Axis Title
0
50
100
150
200
250
300
350
400
450
Axis Title
LES
Filtered DNS


LES VS Filtered DNS Solution of Burgers Equation
(Smagorinsky)
LES :Number of Grids= 101
DNS :Number of Grids= 801
Time=1.3087, Filter size= 8* Delt_x (DNS)
Kinematic Viscosity=0.001
or Re=1000
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
-0.1
0
20
40
60
80
100
120
LES
Filtered DNS